Tutorial 2 - Solutions

1. (a) Can be interpreted as a frequency. *Repeatable expt:* Repeatedly roll the die and record the sequence of score.
   (b) Can be interpreted as a frequency. *Repeatable expt:* Select individual randomly from all 20-yr-old males then record lifespan. Note that this interpretation assumes an infinite pool of people to choose from - not entirely realistic.
   (c) Cannot be interpreted as a frequency. There's only one George Clooney.
   (d) Cannot be interpreted as a frequency.
   (e) Can be interpreted as a frequency if you regard the next pint as a random draw from the population of pints of milk in supermarkets.

2. (a) $\text{Bin}(10, p)$ with some unknown probability p.
   (b) $\text{Geo}\,(p)$ might be plausible if results of throws are independent and probability of success p is constant over throws. Not entirely realistic given p might increase with practice.
   (c) $\text{Poi}(\lambda)$ might be appropriate here.
   (d) A normal distribution: $\text{N}\,(\mu, \sigma^2)$.
   (e) Possibly a $\text{Gamma}\,(\alpha, \beta)$ distribution or a normal distribution would be appropriate.

3. Let $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, then

$$\begin{aligned}
M_{\bar{X}}(\theta) &= \mathbb{E}(e^{\theta\bar{X}}) = \mathbb{E}(e^{\frac{\theta}{n}(X_1+X_2+\ldots+X_n)}) \\
&= \mathbb{E}(e^{\frac{\theta}{n}X_1}) \cdots \mathbb{E}(e^{\frac{\theta}{n}X_n}) \quad \text{(by independence of } X_1, X_2, ..., X_n\text{)} \\
&= \left(\mathbb{E}(e^{\frac{\theta}{n}X_1})\right)^n \quad \text{(since } X_1, X_2, ..., X_n \text{ are identically distributed)} \\
&= \left(M_{X_1}\left(\frac{\theta}{n}\right)\right)^n = \left(\left(\frac{\lambda}{\lambda - \frac{\theta}{n}}\right)^t\right)^n \quad \text{(since } X_1 \sim \text{Gamma}(t, \lambda)\text{)} \\
&= \left(\frac{n\lambda}{n\lambda - \theta}\right)^{nt} \quad \text{(algebra)}.
\end{aligned}$$

   Thus the mgf of $\bar{X}$ corresponds to the mgf for a $\text{Gamma}\,(nt, n\lambda)$ random variable, and by the uniqueness of mgfs, we conclude that $\bar{X} \sim \text{Gamma}\,(nt, n\lambda)$.

4. We have:

$$\begin{aligned}
\text{MSE}(g(\underline{X})) &= \mathbb{E}((g(\underline{X}) - \theta)^2) \\
&= \mathbb{E}((g(\underline{X}))^2) - 2\mathbb{E}(g(\underline{X}))\theta + \theta^2 \\
&= [\mathbb{E}((g(\underline{X}))^2) - (\mathbb{E}(g(\underline{X})))^2] + [(\mathbb{E}(g(\underline{X})))^2 - 2\mathbb{E}(g(\underline{X}))\theta + \theta^2] \\
&= \mathbb{V}\text{ar}(g(\underline{X})) + [\mathbb{E}(g(\underline{X})) - \theta]^2 \\
&= \mathbb{V}\text{ar}(g(\underline{X})) + (\text{Bias}(g(\underline{X})))^2.
\end{aligned}$$

5. (a) $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, so $\mathbb{E}(\bar{X}) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(X_i) = \frac{1}{n}\sum_{i=1}^{n}\mu = \mu$.

(b) Since $X_1, X_2, ..., X_n$ are *independent* and *identically distributed*, we have:

$$\mathbb{V}\text{ar}(\bar{X}) = \frac{1}{n^2}\mathbb{V}\text{ar}(\sum_{i=1}^{n} X_i) = \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{V}\text{ar}(X_i) = \frac{1}{n^2}\sum_{i=1}^{n}\sigma^2 = \frac{\sigma^2}{n}.$$

(c)

$$\begin{aligned} S^2 &= \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \\ &= \frac{1}{n-1}\sum_{i=1}^{n} X_i^2 - \frac{2}{n-1}\bar{X}\sum_{i=1}^{n} X_i + \frac{n}{n-1}(\bar{X})^2 \\ &= \frac{1}{n-1}\sum_{i=1}^{n} X_i^2 - \frac{2n}{n-1}(\bar{X})^2 + \frac{n}{n-1}(\bar{X})^2 \\ &= \frac{1}{n-1}\sum_{i=1}^{n} X_i^2 - \frac{n}{n-1}(\bar{X})^2. \end{aligned}$$

Recall that $\mathbb{E}(X^2) = \mathbb{V}\text{ar}(X) + (\mathbb{E}(X))^2 = \sigma^2 + \mu^2$ and $\mathbb{E}(\bar{X}^2) = \mathbb{V}\text{ar}(\bar{X}) + (\mathbb{E}(\bar{X}))^2 = \frac{\sigma^2}{n} + \mu^2$. So

$$\begin{aligned} \mathbb{E}(S^2) &= \frac{1}{n-1}\sum_{i=1}^{n}\mathbb{E}(X_i^2) - \frac{n}{n-1}\mathbb{E}(\bar{X}^2) \\ &= \frac{n}{n-1}(\sigma^2 + \mu^2) - \frac{n}{n-1}\left(\frac{\sigma^2}{n} + \mu^2\right) \\ &= \sigma^2. \end{aligned}$$

6. (a) Yes. $\mathbb{E}(\hat{\lambda}) = \mathbb{E}(\bar{X}) = \mathbb{E}(X) = \lambda$, therefore $\text{Bias}(\bar{X}) = 0$.

(b) $\text{MSE}(\hat{\lambda}) = \mathbb{V}\text{ar}(\bar{X}) = \frac{\lambda}{n}$, since ($\text{Bias}(\hat{\lambda}) = 0$ and $\mathbb{V}\text{ar}(X) = \lambda$.

(c)

$$\begin{aligned} \text{MSE}(a\hat{\lambda}) = \text{MSE}(a\bar{X}) &= \text{Bias}^2(a\bar{X}) + \mathbb{V}\text{ar}(a\bar{X}) \\ &= (a-1)^2\lambda^2 + \frac{a^2\lambda}{n} \end{aligned}$$

This is a quadratic in a which can easily be shown to be minimised by $a = \frac{n\lambda}{n\lambda+1}$. Note that we would need to know λ in order to obtain the optimal a and that the optimal value of $a \to 1$ as the sample size n increases.

7. (a) Since $\mathbb{E}(S^2) = \sigma^2$ (see Question 5 above), S^2 is an **unbiased** estimator for σ^2. So we have $\text{MSE}(S^2) = \mathbb{V}\text{ar}(S^2) = \frac{2\sigma^4}{n-1}$.

(b) In the *special case* when $\underline{X}$ is a random sample from $N(0, \sigma^2)$, we have $\mathbb{E}(X^2) = \mathbb{V}\text{ar}(X) = \sigma^2$. Thus, another unbiased estimator of σ^2 is given by $S_n^2 = \frac{1}{n}\sum_{i=1}^n X_i^2$, since $\mathbb{E}(S_n^2) = \frac{1}{n}\sum_{i=1}^n \mathbb{E}(X_i^2) = \sigma^2$.

To compute $\text{MSE}(S_n^2) = \mathbb{V}\text{ar}(S_n^2)$, we note that since $X_1, X_2, ..., X_n$ are i.i.d. variables,

$$\mathbb{V}\text{ar}(S_n^2) = \frac{1}{n^2}\sum_{i=1}^n \mathbb{V}\text{ar}(X_i^2) = \frac{1}{n^2}\sum_{i=1}^n 2\sigma^4 = \frac{2\sigma^4}{n}.$$

Note: $\mathbb{V}\text{ar}(X_i^2) = \mathbb{E}(X_i^4) - (\mathbb{E}(X_i^2))^2 = \mathbb{E}(X_i^4) - \sigma^4 = 3\sigma^4 - \sigma^4 = 2\sigma^4$. (Use the mgf for $X_i \sim N(0, \sigma^2)$, to show that $\mathbb{E}(X_i^4) = 3\sigma^4$).

(c) Finally, consider $S_{n+2}^2 = \frac{1}{n+2}\sum_{i=1}^n X_i^2 = \frac{n}{n+2}S_n^2$. We have $\mathbb{E}(S_{n+2}^2) = \frac{n}{n+2}\mathbb{E}(S_n^2) = \frac{n}{n+2}\sigma^2$, so S_{n+2}^2 is biased with

$$\text{Bias}(S_{n+2}^2) = \frac{n}{n+2}\sigma^2 - \sigma^2 = -\frac{2}{n+2}\sigma^2.$$

Next, $\mathbb{V}\text{ar}(S_{n+2}^2) = \mathbb{V}\text{ar}(\frac{n}{n+2}S_n^2) = \frac{n^2}{(n+2)^2}\frac{2\sigma^4}{n}$, so

$$\text{MSE}(S_{n+2}^2) = \mathbb{V}\text{ar}(S_{n+2}^2) + (\text{Bias}(S_{n+2}^2))^2 = \frac{2\sigma^4}{n+2}.$$

Discussion: Both S^2 and S_n^2 are unbiased estimators, with S_n^2 more efficient than S^2. But of the three estimators, S^2, S_n^2, and S_{n+2}^2, it is the *biased* estimator S_{n+2}^2 that is the most efficient.

8. (a) First, determine the mean and variance of the population distribution:

$$\mu = \mathbb{E}(X) = \int_0^\theta x f_X(x;\theta)dx = \int_0^\theta \frac{2x^2}{\theta^2}dx = \frac{2\theta}{3}.$$

$$\mathbb{E}(X^2) = \int_0^\theta x^2 f_X(x;\theta)dx = \int_0^\theta \frac{2x^3}{\theta^2}dx = \frac{\theta^2}{2}.$$

So:

$$\sigma^2 = \mathbb{V}\text{ar}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{\theta^2}{2} - \frac{4\theta^2}{9} = \frac{\theta^2}{18}.$$

Next, using the properties of expectation, we have

$$\mathbb{E}(\hat{\theta}(\underline{X})) = \mathbb{E}\left(\frac{3\bar{X}}{2}\right) = \frac{3}{2}\mathbb{E}(\bar{X}) = \frac{3\mu}{2} = \frac{3}{2}\left(\frac{2\theta}{3}\right) = \theta.$$

So $\hat{\theta}$ is an *unbiased* estimator for θ and

$$\text{MSE}(\hat{\theta}(\underline{X})) = \mathbb{V}\text{ar}(\hat{\theta}(\underline{X})) = \mathbb{V}\text{ar}\left(\frac{3\bar{X}}{2}\right) = \frac{9}{4}\mathbb{V}\text{ar}(\bar{X}) = \frac{9}{4}\frac{\sigma^2}{n} = \frac{\theta^2}{8n}.$$

(b) Suppose that $Y = a\bar{X}$, then

$$\mathbb{E}(Y) = \mathbb{E}(a\bar{X}) = a\mathbb{E}(\bar{X}) = a\mu = \frac{2a\theta}{3}.$$

and

$$\text{Bias}(Y) = \mathbb{E}(Y) - \theta = \frac{2a}{3} - \theta = \left(\frac{2a}{3} - 1\right)\theta.$$

Also,

$$\mathbb{V}\text{ar}(Y) = \mathbb{V}\text{ar}(a\bar{X}) = a^2\mathbb{V}\text{ar}(\bar{X}) = a^2\frac{\sigma^2}{n} = \frac{a^2\theta^2}{18n}.$$

Hence

$$\begin{aligned}\text{MSE}(Y) &= \mathbb{V}\text{ar}(Y) + (\text{Bias}(Y))^2 \\ &= \tfrac{a^2\theta^2}{18n} + \left(\tfrac{2a}{3} - 1\right)^2\theta^2.\end{aligned}$$

To minimize $\text{MSE}(Y)$ with respect to a, we solve

$$\frac{d}{da}\text{MSE}(Y) = \frac{a\theta^2}{9n} + \frac{4}{3}\left(\frac{2a}{3} - 1\right)\theta^2 = 0,$$

i.e. solve

$$\frac{a}{9n} + \frac{8a}{9} - \frac{4}{3} = 0$$

to obtain $a^* = \frac{12n}{1+8n}$. To check that $\text{MSE}(Y)$ is minimised at a^*, we note that

$$\frac{d^2}{da^2}\text{MSE}(Y) = \frac{\theta^2}{9n} + \frac{8\theta^2}{9} > 0.$$

So $\text{MSE}(Y)$ is minimised when $a^* = \frac{12n}{1+8n}$ and the most efficient estimator of the form $Y = a\bar{X}$ is given by $Y^* = \frac{12n}{1+8n}\bar{X}$.

9. (a)

$$\begin{aligned}\mathbb{E}[\bar{X}] &= \mathbb{E}[X] = \theta \\ \mathbb{V}\text{ar}(\bar{X}) &= \frac{1}{n}\mathbb{V}\text{ar}(X) = \frac{\theta^2}{n}.\end{aligned}$$

This is unbiased so

$$\text{MSE}(\bar{X}) = \text{Bias}(\bar{X})^2 + \mathbb{V}\text{ar}(\bar{X}) = \frac{\theta^2}{n}.$$

(b) Let $Y = a\bar{X}$ so

$$\mathbb{E}[Y] = a\mathbb{E}[\bar{X}] = a\theta.$$

So the bias is $\text{Bias}(\bar{X}) = a\theta - \theta = (a-1)\theta$. Also

$$\mathbb{V}\text{ar}(Y) = \mathbb{V}\text{ar}(a\bar{X}) = \frac{a^2}{n}\mathbb{V}\text{ar}(X) = \frac{a^2\theta^2}{n}.$$

Therefore the MSE is $\text{MSE}(Y) = (a-1)^2\theta^2 + \frac{a^2\theta^2}{n}$. We can minimise this by differentiating and equating to zero:

$$\frac{d}{da}\text{MSE}(Y) = 2(a-1)\theta^2 + \frac{2a\theta^2}{n}.$$

Equating this to zero gives the following equation

$$2(a-1)\theta^2 = -\frac{2a\theta^2}{n}$$

Rearranging

$$(2n\theta^2 + 2\theta^2)a = 2n\theta^2 \Rightarrow a = \frac{n}{n+1}.$$

Note that $\text{MSE}(Y)$ is a quadratic in a with the coefficient of a^2 being positive so it has a unique stationary point which is the global minimum.

10. Let $Y = a(n-1)S^2$ then

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[a(n-1)S^2] \\ &= a(n-1)\mathbb{E}[S^2] \\ &= a(n-1)\sigma^2.\end{aligned}$$

Therefore we have

$$\text{Bias}(Y) = a(n-1)\sigma^2 - \sigma^2 = (a(n-1)-1)\sigma^2.$$

Now

$$\begin{aligned}\mathbb{V}\text{ar}(Y) &= \mathbb{V}\text{ar}(a(n-1)S^2 \\ &= a^2(n-1)^2\mathbb{V}\text{ar}(S^2) \\ &= a^2(n-1)^2\frac{2\sigma^4}{n-1} && \text{See Question 7} \\ &= 2a^2(n-1)\sigma^4.\end{aligned}$$

We know that

$$\begin{aligned}\text{MSE}(Y) &= \mathbb{V}\text{ar}(Y) + \text{Bias}(Y)^2 \\ &= 2a^2(n-1)\sigma^4 + (a(n-1)-1)^2\sigma^4.\end{aligned}$$

To minimise let

$$\begin{aligned}F(a) &= 2a^2(n-1)\sigma^4 + (a(n-1)-1)^2\sigma^4 \\ F'(a) &= 4a(n-1)\sigma^4 + 2(a(n-1)-1)(n-1)\sigma^4 \\ &= 2\sigma^4(n-1)(2a + a(n-1) - 1) \\ &= 2\sigma^4(n-1)((n+1)a - 1).\end{aligned}$$

Therefore $F'(a) = 0$ if and only if

$$2\sigma^4(n-1)((n+1)a-1) = 0$$

which occurs if $a = (n+1)^{-1}$.

Checking the second derivative we see that $F''(a) = 2\sigma^4(n^2-1)$ which is positive for $n \geq 1$ so the point $a = (n+1)^{-1}$ is the minimum of F. Hence this value of a gives the most efficient estimator.

11. We have

$$g_1(\underline{X}) = \frac{Y}{8}, \quad g_2(\underline{X}) = \frac{Y+1}{10}, \quad Y = \sum_{i=1}^{8} X_i \sim \text{Bin}(8, p).$$

Since $\mathbb{E}[Y] = 8p$ we have

$$\mathbb{E}[g_1(\underline{X})] = \frac{\mathbb{E}[Y]}{8} = p, \quad \mathbb{E}[g_2(\underline{X})] = \frac{\mathbb{E}[Y]+1}{10} = \frac{8p+1}{10}.$$

So $g_1(\underline{X})$ is unbiased and $g_2(\underline{X})$ is biased. Now using that $\mathbb{V}\text{ar}(Y) = 8p(1-p)$ we have

$$\mathbb{V}\text{ar}[g_1(\underline{X})] = \frac{\mathbb{V}\text{ar}[Y]}{8^2} = \frac{p(1-p)}{8}, \quad \mathbb{V}\text{ar}[g_2(\underline{X})] = \frac{\mathbb{V}\text{ar}[Y]}{10^2} = \frac{8p(1-p)}{10^2}.$$

Combining the variance and bias we can calculate the MSE

$$\begin{aligned}
\text{MSE}(g_1(\underline{X}) &= \text{Bias}(g_1(\underline{X}))^2 + \mathbb{V}\text{ar}(g_1(\underline{X})) = \frac{p(1-p)}{8}. \\
\text{MSE}(g_2(\underline{X}) &= \text{Bias}(g_2(\underline{X}))^2 + \mathbb{V}\text{ar}(g_2(\underline{X})) = \left(\frac{8p+1}{10} - p\right)^2 + \frac{8p(1-p)}{10^2} \\
&= \frac{(1-2p)^2}{100} + \frac{8}{100}p(1-p).
\end{aligned}$$

From the plot we see that $g_1(\underline{X})$ is more efficient near $p = 0$ and $p = 1$, to find the range of values we need to first determine the values of p where $\text{MSE}(g_1(\underline{X})) = \text{MSE}(g_2(\underline{X}))$. This occurs when

$$\frac{p(1-p)}{8} = \frac{(1-2p)^2}{100} + \frac{8}{100}p(1-p)$$

Rearranging this gives

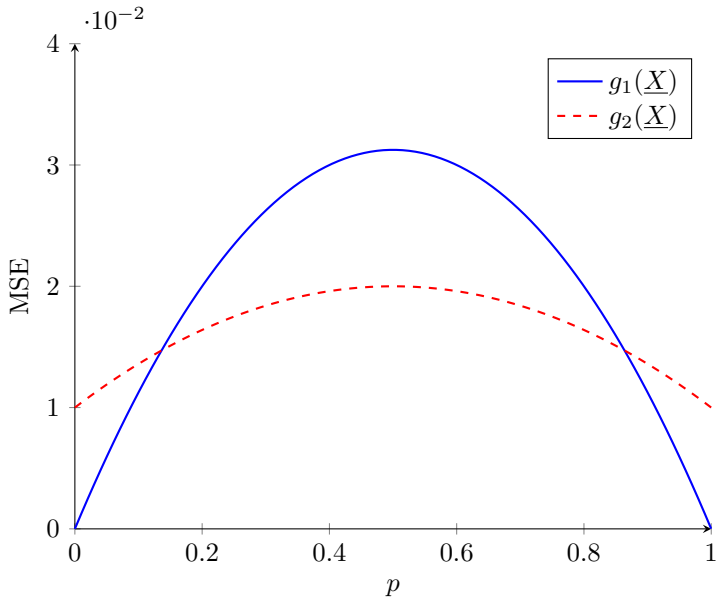
$$0 = 1 - \frac{17}{2}p + \frac{17}{2}p^2.$$

$$0 = 1 - \frac{17}{2}p + 172p^2.$$

We can solve this using the quadratic formula

$$p = \frac{\frac{17}{2} \pm \sqrt{\frac{17^2}{2^2} - 2(17)}}{17} = \frac{1}{2} \pm \frac{3}{2\sqrt{17}}$$

So $g_1(\underline{X})$ is more efficient for $p \leq \frac{1}{2} - \frac{3}{2\sqrt{17}} \approx 0.136$ or for $p > \frac{1}{2} + \frac{3}{2\sqrt{17}} \approx 0.864$.



Exam Style Question

(a) Suppose that $\underline{X} = (X_1, X_2, \ldots, X_n)$ is a random sample from a distribution parametrised by unknown θ and let $g(\underline{X})$ denote an estimator for θ.

(i) The bias of $g(\underline{X})$ is

$$\text{Bias}(g(\underline{X})) = \mathbb{E}[g(\underline{X})] - \theta.$$

The mean square error of $g(\underline{X})$ is

$$\text{MSE}(g(\underline{X})) = \mathbb{E}[(g(\underline{X}) - \theta)^2].$$

[2 marks]

(ii)

$$\text{MSE}(g(\underline{X})) = \text{Bias}(g(\underline{X}))^2 + \mathbb{V}\text{ar}(g(\underline{X})).$$

[1 marks]

(b) Let $\underline{X}$ be a random sample of size n from a distribution with density function

$$f_X(x;\theta) = \begin{cases} \frac{1}{\theta^2} x e^{-\frac{x}{\theta}}, & x > 0 \\ 0, & x \leq 0. \end{cases}$$

Here $\theta > 0$ is an unknown parameter. Note that this corresponds to a Gamma $\left(2, \frac{1}{\theta}\right)$ distribution and has mean 2θ and variance $2\theta^2$.

(i) We will use the expression for the mean square error given in (a)(ii). First we find the bias:

$$\begin{aligned} \text{Bias}(\hat{\theta}_1(\underline{X})) &= \mathbb{E}\left[\frac{1}{2}\bar{X}\right] - \theta \\ &= \frac{1}{2}\mathbb{E}\left[\bar{X}\right] - \theta \end{aligned}$$

Since the sample mean is an unbiased estimator for the mean of X we have $\mathbb{E}\left[\bar{X}\right] = \mathbb{E}\left[X\right]$. We are given that $X \sim \text{Gamma}\left(2, \frac{1}{\theta}\right)$ so $\mathbb{E}[X] = 2\theta$. Therefore

$$\text{Bias}(\hat{\theta}_1(\underline{X})) = \frac{1}{2}\mathbb{E}\left[\bar{X}\right] - \theta\frac{1}{2}(2\theta) - \theta = 0.$$

That is, $\hat{\theta}_1$ is unbiased.
Now for the variance:

$$\mathbb{V}\text{ar}(\hat{\theta}_1(\underline{X})) = \mathbb{V}\text{ar}\left(\frac{1}{2}\bar{X}\right) = \frac{1}{4}\mathbb{V}\text{ar}(\bar{X}).$$

Note that

$$\mathbb{V}\text{ar}(\bar{X}) = \frac{1}{n}\mathbb{V}\text{ar}(X) = \frac{2\theta^2}{n}.$$

Hence

$$\mathbb{V}\text{ar}(\hat{\theta}_1(\underline{X})) = \frac{\theta^2}{2n}.$$

Now combining these terms we have

$$\text{MSE}(\hat{\theta}_1(\underline{X})) = \frac{\theta^2}{2n}.$$

Observe that as $n \to \infty$ the mean square error $\text{MSE}(\hat{\theta}_1(\underline{X})) \to 0$.

[4 marks]

(ii) Note that

$$\begin{aligned}
\mathbb{E}[\hat{\theta}_2(\underline{X})] &= c\mathbb{E}[\bar{X}] = 2c\theta, \\
\text{Bias}(\hat{\theta}_2(\underline{X})) &= (2c-1)\theta, \\
\mathbb{V}\text{ar}(\hat{\theta}_2(\underline{X})) &= c^2\mathbb{V}\text{ar}(\bar{X}) = \frac{2c^2\theta^2}{n}.
\end{aligned}$$

Hence

$$\text{MSE}(\hat{\theta}_2(\underline{X})) = \text{Bias}(\hat{\theta}_2(\underline{X}))^2 + \mathbb{V}\text{ar}(\hat{\theta}_2(\underline{X})) = (2c-1)^2\theta^2 + \frac{2c^2\theta^2}{n}$$

[4 marks]

(iii) To find the minimum we will differentiate this expression for the mean square error with respect to c

$$\frac{d}{dc}\left((2c-1)^2\theta^2 + \frac{2c^2\theta^2}{n}\right) = 4(2c-1)\theta^2 + \frac{4c\theta^2}{n}.$$

Find the value c^* where $\frac{d}{dc}\text{MSE}(\hat{\theta}_2(\underline{X})) = 0$ as

$$c^* = \frac{n}{2n+1}.$$

[3 marks]